

Voice Messaging System Project Documentation

Abstract

This project is a cloud-based voice messaging system that allows users to record audio through a browser, upload it to a backend server, store it in cloud storage, and send real-time SMS notifications with access links. The system removes dependency on traditional phone call voicemail systems and provides a fully web-based solution.

Tech Stack

Frontend	HTML, JavaScript, MediaRecorder API
Backend	Python, Flask
Database & Storage	Supabase (PostgreSQL + Storage)
Messaging	Twilio SMS
Tunneling	ngrok

Features Included

- Browser-based voice recording - Cloud storage of audio files - Database storage of metadata - SMS notification with voice link - Real-time backend processing

Workflow

User opens web app → Records voice → Audio sent to Flask backend → Stored in Supabase → SMS sent via Twilio → User receives link → Plays voice message.

System Architecture

Frontend (Browser UI) ↓ Flask Backend API ↓ Supabase Storage + Database ↓ Twilio SMS Notification ↓ User Access via Link